

State-Conditional Adversarial Learning: An Off-Policy Visual Domain Transfer Method for End-to-End Imitation Learning

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Abstract: We study visual domain transfer for end-to-end imitation learning in a realistic and challenging setting where target-domain data are strictly off-policy, expert-free, and scarce. We first provide a theoretical analysis showing that the target-domain imitation loss can be upper bounded by the source-domain loss plus a state-conditional latent KL divergence between source and target observation models. Guided by this result, we propose State-Conditional Adversarial Learning (SCAL), an off-policy adversarial framework that aligns latent distributions conditioned on system state using a discriminator-based estimator of the conditional KL term. Experiments on visually diverse autonomous driving environments built on the BARC-CARLA simulator demonstrate that SCAL achieves robust transfer and strong sample efficiency.

Keywords: Domain Transfer, Imitation Learning, Few-shot Transfer Learning, Data-driven Control

1. INTRODUCTION

Vision-based imitation learning (IL) has achieved impressive results across diverse robotic domains, including aerial drones Xing et al. (2024); Pfeiffer et al. (2022), autonomous driving Bojarski et al. (2016), and manipulation Rahmatizadeh et al. (2017); Zhu et al. (2022). By learning directly from high-dimensional visual observations, these methods avoid reliance on specialized sensors while enabling agents to reproduce complex expert behaviors. However, despite their empirical success, vision-based IL policies remain brittle when deployed in visual domains insufficiently represented in the training distribution.

Existing approaches for improving generalization in vision-based IL can be broadly categorized into zero-shot and few-shot adaptation methods. Zero-shot techniques such as domain randomization Tobin et al. (2017); Mehta et al. (2020), DARLA Higgins et al. (2017), and DARL Li et al. (2021) attempt to transfer policies without any access to target-domain data, relying on the assumption that synthetic variations can capture target-domain visual characteristics. Few-shot methods, by contrast, leverage limited target-domain data Bewley et al. (2018); Zhou et al. (2021); Stadie et al. (2017a,b); Hansen et al. (2021), but typically impose strong assumptions on what information is available. Several works assume the agent may execute on-policy rollouts in the target domain to gather online data Zhou et al. (2021); Stadie et al. (2017a); Hansen et al. (2021). Others assume access to target-

domain expert demonstrations Sun et al. (2019). Although Bewley et al. (2018) removes the need for both rollouts and expert demonstrations, its CycleGAN-based pixel translation requires large unlabeled target datasets, making it unsuitable for data-scarce settings.

In many real-world scenarios, these assumptions do not hold. Target environments are often safety-critical or operationally expensive, making on-policy exploration costly or infeasible. Human or controller-based demonstrations may be difficult to obtain due to labor constraints, hardware wear, or lack of reliable expert solutions. Even passive data collection is limited by hardware availability, mission time, or regulatory restrictions.

Motivated by these practical constraints, we formalize a more realistic and challenging few-shot adaptation setting characterized as follows:

- The system cannot interact with the target environment directly during training, but a small data set is available, without expert supervision. Note that the data may not follow the distribution induced by the policy.
- The system can interact with a similar environment during training, and an expert is available to provide supervision for the policy.

This specific setting, to our knowledge, is not adequately addressed by existing few-shot IL domain adaptation frameworks, and we leverage the information made available in this setting to achieve better performance and data efficiency. We provide a surrogate upper bound of the imitation loss, without access to the expert supervision in the target environment, and empirically show that under certain conditions, optimizing this surrogate is a data-

* The codes for our implementation is available at <https://github.com/Xiang-Foothill/BkgGeneralizor.git>

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efficient approach to recover a high-performing vision-based policy with off-policy data.

The main contribution of this paper is as follows:

- We provide a formal analysis of vision-based imitation learning under domain shift, deriving upper bounds on target-domain performance in terms of conditional latent divergence.
- Guided by this theory, we introduce State-Conditional Adversarial Learning (SCAL), a novel off-policy, expert-free domain-transfer framework that aligns conditional latent distributions using only a small, off-policy target-domain dataset paired with state information.
- We empirically validate our theoretical insights in challenging visual driving environments, demonstrating that our method achieves significantly improved sample efficiency compared to existing approaches while operating under more restrictive and realistic assumptions.

2. PROBLEM FORMULATION

2.1 Definition and Control Objective

Consider the following class of non-linear, time-invariant, deterministic, discrete-time system with stochastic observations:

$$\begin{aligned} x_{k+1} &= f(x_k, u_k), \quad y_k \sim e(\cdot | x_k), \quad x_0 \sim \mathcal{X}_0, \\ x_k &\in \mathcal{X}, \quad u_k \in \mathcal{U}, \quad y_k \in \mathcal{Y}, \quad \forall k \in \mathbb{N}. \end{aligned} \quad (1)$$

where x_k, u_k are the state and input at time k ; f is the dynamics, which is assumed to be known; y_k is the observation at time k , which follows the unknown state-dependent observation distribution $e(\cdot | x_k)$; $\mathcal{X}_0$ is the initial state distribution; $\mathcal{X}, \mathcal{U}$, and $\mathcal{Y}$ are the state space, the action space, and the observation space, respectively.

Let $\pi_\theta : \mathcal{Y} \mapsto \mathcal{U}$ be a parametric observation feedback policy. Following standard end-to-end visuomotor imitation learning Levine et al. (2016); Bojarski et al. (2016); Zhou et al. (2021), we parameterize the policy as $\pi_\theta(y_k) = D_w(E_\phi(y_k))$, where E_ϕ is a parametric encoder network, D_w is the parametric control head, and

$$l_k = E_\phi(y_k) \in \mathcal{L}$$

is the latent representation of the observation y_k , and $\mathcal{L}$ denotes the latent space. $\theta = [w \ \phi]$ is the collection of all parameters of the policy.

The interaction between any agent π and system (1) can be viewed as a Markov decision process (MDP). Let $p^k(\cdot | e, \pi)$ be state distribution of such MDP at k -th step.

Definition 2.1 (Discounted Visitation Distribution (Ho and Ermon (2016); Kakade and Langford (2002))). *Let $\gamma \in (0, 1)$ be a discount factor. The discounted state visitation distribution induced by policy π is defined as:*

$$p(x | e, \pi) \triangleq (1 - \gamma) \sum_{k=0}^{\infty} \gamma^k p^k(x_k = x | e, \pi). \quad (2)$$

For a given encoder E , the latent distribution conditioned at a given state x is:

$$p(l | x, e, E) = \Pr(l = E(y) \mid y \sim e(\cdot | x)) \quad (3)$$

We will refer to the above distribution a state-conditioned latent distribution.

Suppose we have two systems sharing the same known dynamics f but distinct observation models e_s and e_t , where e_s is known and e_t is unknown. We refer to the e_s as the **source domain**, and e_t as the **target domain**. As a shorthand, the domain-specific discounted visitation distributions are denoted as $p_s(\cdot | \pi) = p(\cdot | e_s, \pi)$ and $p_t(\cdot | \pi) = p(\cdot | e_t, \pi)$ for the source domain and the target domain, respectively. Analogously, the same short-hand applies for the state-conditioned latent distribution $p_s(l | x, E) = p(l | x, e_s, E)$ and $p_t(l | x, E) = p(l | x, e_t, E)$. For both domains $e_i \ i \in \{s, t\}$ Let

$$J(\pi; e_i) \triangleq \sum_{k=0}^{\infty} \mathbb{E} [c(x_k, \pi(y_k))], \quad x_k \sim p^k(\cdot | e_i, \pi) \quad y_k \sim e_i(\cdot | x_k). \quad (4)$$

subject to the constraints and distributions in (1). $c(\cdot, \cdot)$ is a non-negative cost function. Let $c(x_k, u_k) = \infty$ if $x_k \notin \mathcal{X}$ or $u_k \notin \mathcal{U}$.

The overall optimization objective is:

$$\min_{\theta} \frac{1}{2} (J(\pi_\theta; e_s) + J(\pi_\theta; e_t)) \quad (5)$$

2.2 Supervision

We assume a high-performing black-box expert π_β which provides supervision u_k^* at a given state x_k ,

$$u_k^* = \pi_\beta(x_k, h_k),$$

where $h_k = h(\mathbf{x}_{0:k-1})$ is the hidden state of the expert, which contains information of the closed-loop trajectory prior to time k , denoted as $\mathbf{x}_{0:k-1}$.

Since the dynamics f and the source domain observation distribution e_s is known, we can follow the learning framework in DAGger (Ross et al. (2011)) to collect a dataset $\mathcal{B}_s = \{(y_k, x_k, u_k^*)\}$ with asymptotically no covariate shift.

Note that such direct data collection is impossible for the target domain because e_t is unknown. However, we assume access to a small dataset $\mathcal{B}_t = \{(y_k, x_k)\}$ with observation-state pairs. The dataset does not necessarily follow a closed-loop trajectory, so we cannot acquire expert supervision because of the lack of hidden states. In addition, even if supervision is available, vanilla imitation learning framework still suffers from the covariate shift.

2.3 Optimal Control via Imitation Learning

With the expert supervision u_k^* , we can reformulate the objective (5) in the following imitation loss form:

$$\min_{\theta} \mathcal{J}_s(\theta) + \mathcal{J}_t(\theta). \quad (6)$$

where

$$\mathcal{J}_i(\theta) = \mathbb{E}_{(y, u^*) \sim p(\cdot | e_i, \pi_\theta)} [d(\pi_\theta(y), u^*)], \quad i \in \{s, t\} \quad (7)$$

Function $d : \mathcal{U} \times \mathcal{U} \rightarrow \mathbb{R}$ measures action difference. $\mathcal{J}_s(\theta)$ and $\mathcal{J}_t(\theta)$ are imitation surrogates for $J(\pi_\theta; e_s)$, $J(\pi_\theta; e_t)$ in (5) respectively.

Note that (6) cannot be directly solved under imitation learning frameworks because e_t is unknown, and consequently, we lack the data to estimate $\mathcal{J}_t(\theta)$ directly. In

Section 4-5, we leverage adversarial learning to provide an upper bound for $\mathcal{J}_t(\theta)$ as its surrogate, and in Section 6, we empirically show the validity of this approach.

3. RELATED WORKS

3.1 Imitation Learning

Imitation Learning has seen great success in recent years Bojarski et al. (2016); Xing et al. (2024). One important challenge is distributional shift: the trajectory distribution induced by the agent during inference time is not consistent with the trajectory of the expert from the data buffer. DAgger-style framework Ross et al. (2011); Zhang and Cho (2017); Ross and Bagnell (2014) solves this problem by mixing agents' actions with expert actions when collecting data. In this paper, we leverage DAgger as the base learning pipeline in the source-domain.

3.2 Adversarial Learning

Adversarial learning is widely used for distribution alignment by the community of computer vision Arjovsky et al. (2017); Ganin and Lempitsky (2016); Arora et al. (2018). Built upon GAN-style framework, conditional adversarial learning is further introduced to align conditional distributions Long et al. (2018) Miyato and Koyama (2018) Mirza and Osindero (2014) to further tackle the multi-modal cases. In this work, we followed the paradigm of conditional adversarial learning to accomplish the matching between latent distributions conditioned on task-relevant information.

4. TRANSFER LEARNING VIA ALIGNMENT

Imitation learning for an observation-feedback policy requires a dataset of observation-demonstration pairs. While we can sample trajectories in the state space with the known dynamics f , we do not have access to the corresponding observations y_k because e_t is unknown.

The key idea is that while $\mathcal{B}_t$ only contains off-policy data without supervision, we can leverage $\mathcal{B}_s$ and $\mathcal{B}_t$ to establish the alignment between the two observation distributions as a means of transfer learning. In this section, we formally define alignment, and analyze its connection to imitation loss.

4.1 Alignment and Connection to Imitation Learning

Definition 4.1 (Alignment). *Recall the notion of state-conditioned latent distribution in Definition 2.1. If the encoder network E_ϕ induces identical state-conditioned latent distribution under the two observation distributions e_s and e_t for all possible states x i.e.,*

$$p_s(l \mid x, E_\phi) = p_t(l \mid x, E_\phi), \forall x \in \mathcal{X}, l \in \mathcal{L}$$

then e_s and e_t are **aligned** by E_ϕ .

Lemma 4.1. *If the policy π_θ has an encoder E_ϕ that aligns e_s and e_t , then*

$$\mathcal{J}_t(\theta) = \mathcal{J}_s(\theta).$$

Proof. See proof appendix $\square$

Lemma 4.1 implies that under perfect alignment, the performance of π_θ in the source domain exactly matches that in the target domain.

Perfect alignment is challenging to attain, especially with a limited $\mathcal{B}_t$. We quantify the alignment loss of between e_s and e_t as the expected Kullback–Leibler divergence between the distribution of the latent encoded by E_ϕ . Formally,

$$L(\phi) = \mathbb{E}_{p_s(x|\pi_\theta)} [d_{\text{KL}}(p_s(l \mid \pi_\theta) \| p_t(l \mid \pi_\theta))], \quad (8)$$

where $d_{\text{KL}}(\cdot \| \cdot)$ denotes the Kullback–Leibler divergence.

In general, the connection between alignment and imitation loss is as follows.

Theorem 4.1. *For a policy π_θ with visual encoder E_ϕ , its target-domain imitation loss can be upper bounded by*

$$\mathcal{J}_t(\theta) \leq \mathcal{J}_s(\theta) + \alpha \sqrt{\frac{2\gamma}{1-\gamma} (L(\phi) + \sigma)} \quad (9)$$

where

- $\sigma = d_{\text{KL}}(e_s(\cdot \mid x) \| e_t(\cdot \mid x)) \geq 0$,
- $\alpha = \sup_{y \in \mathcal{Y}, u^* \in \mathcal{U}} d(\pi_\theta(y), u^*) \geq 0$ is the uniform bound over the loss function

Proof. See proof appendix. $\square$

Remark. *The term $\frac{2\gamma}{1-\gamma}$ implies that the upper bound will become more conservative if the state visitation distribution is defined with larger γ . σ is a constant describing the difference between e_s and e_t . This implies that e_s and e_t must be meaningfully similar for the upper bound to be reasonably tight.*

Theorem 4.1 shows that the target domain imitation loss $\mathcal{J}_t(\theta)$ can be optimized by minimizing the source domain imitation loss $\mathcal{J}_s$ and the alignment loss $L(\phi)$, without access to on-policy data in the target domain.

Proposition 4.1. *For the sake of Majorization–Minimization, we propose the following as a surrogate for the joint objective in (6):*

$$\min_{\theta} \mathcal{J}_s(\theta) + L(\phi). \quad (10)$$

5. PROPOSED APPROACH

In this section, we propose minimizing the alignment loss $L(\phi)$ in (8) via adversarial learning, and present State-Conditional Adversarial Learning (SCAL), which is an augmentation of traditional imitation learning frameworks for transfer learning through alignment.

5.1 Discriminator-based Off-policy Evaluation

In practice, we estimate it by training a discriminator Q_ψ parameterized with ψ to distinguish between (l, x) pairs sampled from $\mathcal{B}_s$ and $\mathcal{B}_t$. Note that we can sample (l, x) pairs from the data buffer by first sampling (y, x) pairs and

then applying the visual encoder E of π_θ to observations. The optimization of Q_ψ can be framed as follow:

$$\psi^* = \arg \min_{\psi} \frac{1}{\|\mathcal{B}_t\|} \left\{ \sum_{(y,x) \sim \mathcal{B}_t} \log(1 - Q_\psi(E_\phi(y), x)) + \frac{1}{\|\mathcal{B}_s\|} \sum_{(y,x) \sim \mathcal{B}_s} \log Q_\psi(E_\phi(y), x) \right\} \quad (11)$$

Proposition 5.1. *Given a discriminator Q_ψ trained based on (11), we have*

$$\mathbb{E}_{p_s(x|\pi_\theta)} [d_{\text{KL}}(p_s(l|x, \pi_\theta) \| p_t(l|x, \pi_\theta))] \approx \frac{1}{\|\mathcal{B}_s\|} \sum_{(y,x) \sim \mathcal{B}_s} \log \frac{Q_{\psi^*}(E_\phi(y), x) \widehat{p_{\mathcal{B}_t}}(x)}{1 - Q_{\psi^*}(E_\phi(y), x) \widehat{p_{\mathcal{B}_s}}(x)}$$

where $\widehat{p_{\mathcal{B}_t}}(x)$ and $\widehat{p_{\mathcal{B}_s}}(x)$ are some functions approximating the distributions of x in $\mathcal{B}_s$ and $\mathcal{B}_t$

To understand why this proposition is algorithmically sensible, recall that a discriminator trained following the (11) is approximating:

$$Q_{\psi^*}(l, x) \approx \frac{p_{\mathcal{B}_s}(l, x)}{p_{\mathcal{B}_s}(l, x) + p_{\mathcal{B}_t}(l, x)}$$

where $p_{\mathcal{B}_s}(l, x)$ and $p_{\mathcal{B}_t}(l, x)$ are the distributions underlying the data buffers $\mathcal{B}_s$ and $\mathcal{B}_t$ respectively Goodfellow et al. (2014). Then we can have the following derivation:

$$\begin{aligned} & \frac{1}{\|\mathcal{B}_s\|} \sum_{(y,x) \sim \mathcal{B}_s} \log \frac{Q_{\psi^*}(E_\phi(y), x) \widehat{p_{\mathcal{B}_t}}(x)}{1 - Q_{\psi^*}(E_\phi(y), x) \widehat{p_{\mathcal{B}_s}}(x)} \\ & \approx \mathbb{E}_{p_{\mathcal{B}_s}(l,x)} [\log \frac{Q_{\psi^*}(l, x) \widehat{p_{\mathcal{B}_t}}(x)}{1 - Q_{\psi^*}(l, x) \widehat{p_{\mathcal{B}_s}}(x)}] \\ & \approx \mathbb{E}_{p_{\mathcal{B}_s}(l,x)} [\log \frac{p_{\mathcal{B}_s}(l, x) \widehat{p_{\mathcal{B}_t}}(x)}{p_{\mathcal{B}_t}(l, x) \widehat{p_{\mathcal{B}_s}}(x)}] \\ & \approx \mathbb{E}_{p_s(l,x|\pi_\theta)} [\log \frac{p_s(l|x, \pi_\theta)}{p_t(l|x, \pi_\theta)}] \\ & = \mathbb{E}_{p_s(x|\pi_\theta)} [d_{\text{KL}}(p_s(l|x, \pi_\theta) \| p_t(l|x, \pi_\theta))] \end{aligned}$$

One important assumption for the above derivation to hold is that $p_{\mathcal{B}_s}$ can approximate the agent's on-policy distribution p_s , which is sensible under the settings with unlimited access to the source domain system.

In practice, Q_ψ is implemented as a two-layer neural network with l and x concatenated as inputs. $\widehat{p_{\mathcal{B}_t}}(x)$ and $\widehat{p_{\mathcal{B}_s}}(x)$ are implemented as two independent Gaussian Kernel Estimators fitted with data from $\mathcal{B}_t$ and $\mathcal{B}_s$ respectively.

5.2 Adversarial Learning For Policy Improvements

We now introduce State-Conditional Adversarial Learning (SCAL) to solve for objective (6). Per propositions 4.1 and 5.1, the original intractable objective (6) can be optimized by

Algorithm 1 State-Conditional Adversarial Learning

Require: Source buffer $\mathcal{B}_s$, target buffer $\mathcal{B}_t$, initial parameters θ , trade-off λ

- 1: Fit $\widehat{p_{\mathcal{B}_s}}(x)$ and $\widehat{p_{\mathcal{B}_t}}(x)$ using the x -marginals of $\mathcal{B}_s$ and $\mathcal{B}_t$
- 2: **while** not converged **do**
- 3: // **Update discriminator** Q_ψ
- 4: **for** $k = 1, \dots, K_{\text{disc}}$ **do**
- 5: Sample minibatch $\{(y_i^t, x_i^t)\}$ from $\mathcal{B}_t$ and $\{(y_j^s, x_j^s)\}$ from $\mathcal{B}_s$
- 6: Compute latents $l_i^t = E_\phi(y_i^t)$ and $l_j^s = E_\phi(y_j^s)$
- 7: Compute discriminator loss $\mathcal{J}_{\text{adv}} = -\frac{1}{|\mathcal{B}_t|} \sum_i \log(1 - Q_\psi(l_i^t, x_i^t)) - \frac{1}{|\mathcal{B}_s|} \sum_j \log Q_\psi(l_j^s, x_j^s)$
- 8: Update $\psi \leftarrow \psi - \eta_\psi \nabla_\psi \mathcal{J}_{\text{adv}}$
- 9: **end for**
- 10: // **Update policy and encoder** (θ)
- 11: Fill $\mathcal{B}_s$ following DAgger pipeline
- 12: Sample minibatch $\{(y_j^s, u_j^*, x_j^s)\}$ from $\mathcal{B}_s$
- 13: Compute tractable source-domain loss $\mathcal{J}_s$ based on the base RL or IL pipeline
- 14: For each (y_j^s, x_j^s) , compute $l_j^s = E_\phi(y_j^s)$ and $w_j = \log \frac{Q_\psi(l_j^s, x_j^s) \widehat{p_{\mathcal{B}_t}}(x_j^s)}{1 - Q_\psi(l_j^s, x_j^s) \widehat{p_{\mathcal{B}_s}}(x_j^s)}$
- 15: Set adversarial loss $\mathcal{J}_{\text{adv}} = \frac{1}{B} \sum_j w_j$
- 16: Form total loss $\mathcal{J}_{\text{total}} = \mathcal{J}_s + \lambda \mathcal{J}_{\text{adv}}$
- 17: Update $\theta \leftarrow \theta - \eta_\theta \nabla_\theta \mathcal{J}_{\text{total}}$
- 18: **end while**

$$\theta^* = \arg \min_{\theta} \{ \mathcal{J}_s(\theta) + \lambda \mathcal{J}_{\text{adv}}(\theta) \}$$

where $\mathcal{J}_s(\theta) = \underbrace{\mathbb{E}_{(y,x) \sim \mathcal{B}_s(y, u^*)} [\mathcal{J}(\pi_\theta(y), u^*)]}_{\text{source-domain on-policy loss}}$,

$$\mathcal{J}_{\text{adv}}(\theta) = \underbrace{\frac{1}{\|\mathcal{B}_s\|} \sum_{(y,x) \sim \mathcal{B}_s} \log \frac{Q_{\psi^*}(E_\phi(y), x) \widehat{p_{\mathcal{B}_t}}(x)}{1 - Q_{\psi^*}(E_\phi(y), x) \widehat{p_{\mathcal{B}_s}}(x)}}_{\text{domain confusion loss}},$$

where Q_{ψ^*} is optimized based on 11. Note that $\mathcal{J}_s(\theta)$ can be redefined following other more advanced IL pipelines using the data from $\mathcal{B}_s$ or recollecting data from the source domain. Following the convention of adversarial training, the discriminator and the agent in our implementation are trained iteratively to preserve the expressivity of the discriminator.

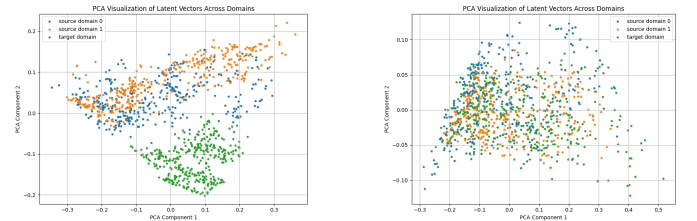


Fig. 1. PCA Visualization of Latent Space with (left) and without(right) using SCAL. The latent vectors presented are sampled from exactly the same path-tracking trajectory.

5.3 Comparison with Prior Works

Compared to Bewley et al. (2018) Higgins et al. (2017), which relies on pixel-level CycleGAN translation and assumes a large pool of unlabeled target images, our framework can tackle realistic settings requiring high sample efficiency. Moreover, our method provides explicit upper bounds on the target-domain loss via conditional latent KL divergence, offering theoretical guarantees absent in purely generative translation approaches. Hansen et al. (2021) Zhou et al. (2021) introduces inverse dynamics as self-supervision signal for domain-adaptation. However, their framework requires on-policy target-domain data whereas our approach restricts target-domain data to be off-policy. Sun et al. (2019) offers a complete theoretical framework for IL visual domain transfer. Comparing to our expert-free assumption, their method assumes optimal expert demonstrations from the target domain. To the best of our knowledge, our framework is the first one that tackles visual domain adaptation under expert-free, off-policy target-domain data assumption while maintaining high sample efficiency.

6. EXPERIMENT

In this section, our empirical analysis aims to verify: (A) the validity of the optimization surrogate in proposed by our theoretical analysis (B) The consistent sample efficiency of our method under various state distribution shifts. What’s more, to outline the practical significance of our approach, we attach an impressive low-speed-to-high-speed transfer learning experiment.

We base all of our experiment designs on Berkeley Autonomous Racing Car simulation environment (BARC), which is a variation of Carla gym. The source-domain system and target domain systems are two carla-environments with the same track shape but drastically different visual appearances.

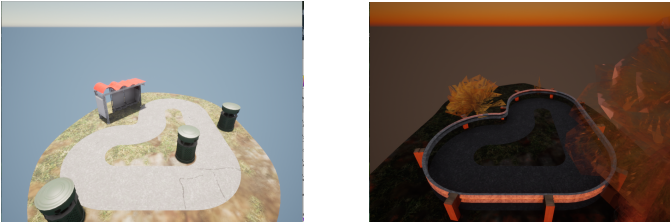


Fig. 2. Two Example Domains in our experiments with the same track shape but drastically different visual characters.

6.1 Off-Policy Evaluation Study

We verify the validity of our theoretical analysis by presenting the strong positive correlation between $\mathcal{J}_t(\theta)$ (the intractable term in objective (6)) and the quantity

$$\mathcal{J}_s(\theta) + \mathbb{E}_{p_s(x|\pi_\theta)}[d_{\text{KL}}(p_s(l|x, \pi_\theta) || p_t(l|x, \pi_\theta))]$$

(the surrogate proposed by our theoretical analysis).

We prepare 20 agents ($\pi_{\theta_0}, \pi_{\theta_1}, \dots, \pi_{\theta_{19}}$), each trained from one unique source domain until $\mathcal{J}_s(\theta_i) \approx 0 \forall i$. We set the

expert of all these agents to be the same black-boxed PID controller without knowing its inner implementation. The value of the State-Conditional KL-divergence

$$\mathbb{E}_{p_s(x|\pi_{\theta_i})}[d_{\text{KL}}(p_s(l|x, \pi_{\theta_i}) || p_t(l|x, \pi_{\theta_i}))]$$

is then calculated based on proposition 5.1 for each of these agents. Then, without any further training, the $\mathcal{J}_t(\theta_i)$ is estimated from two dimensions in the target domain: on-policy imitation loss and on-policy trajectory length.

As illustrated by figure 3, given that each agent has almost perfect source domain on-policy loss, there is a strong correlation between the estimated state-conditional KL divergence and the agent’s on-policy behavior in the target domain.

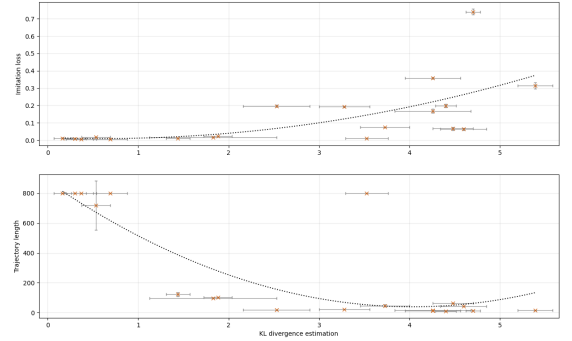


Fig. 3. Correlation between estimated State-Conditional KL divergence and On-policy target domain metric.

6.2 Distributional-shift Study

We are interested in the sample efficiency of our framework under different distributions of $\mathcal{B}_t$. We choose π_β to be a path tracking PID controller. We predefine 3 different state space distributions to collect $\mathcal{B}_t$. Following these state space distributions, we collected $\mathcal{B}_s$ with varying sizes 2048, 1024, 512, 256, 213, 170, 128. For each $\mathcal{B}_t$, five independent trials of training are conducted following our pipeline, and we record the maximum length achieved by the agent in the target domain.

As a performance reference, we include DAGger Ross et al. (2011) with perfect information availability assumption in the target domain. More specifically, we directly train this baseline under a fully supervised and online setting with access to expert demonstrations in the target domain. Under superior information availability from the target domain, this idealized baseline should give an upper bound for the sample efficiency of all off-policy transfer learning algorithms.

As illustrated by figure 5, despite relying solely on an offline buffer without expert supervision, our approach achieves comparable or superior trajectory lengths to DAGger under all three off-policy $\mathcal{B}_t$ distributions. Notably, SCAL maintains strong performance and stability even in low-data regimes (e.g., with only 256 target samples), demonstrating its competitive sample efficiency. What’s more, the experimental result also proves the robustness of our methods to various distributions $\mathcal{B}_t$.

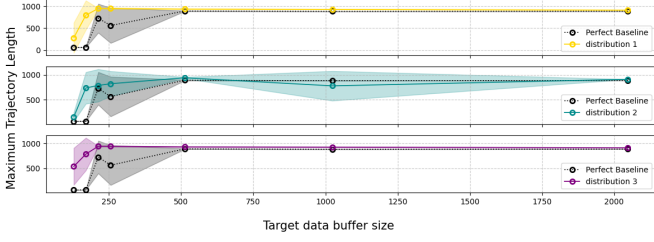


Fig. 4. SCAL compared with perfect baseline under different $\mathcal{B}_s$ distributions. x-axis: Target-domain buffer size. y-axis: Maximum trajectory length achieved in the target domain. SCAL trained with $\mathcal{B}_t$ distribution 1 (yellow); SCAL trained with $\mathcal{B}_t$ distribution 2 (blue); SCAL trained with $\mathcal{B}_t$ distribution 3 (purple). Perfect baseline (Black). The shaded area represents variance.

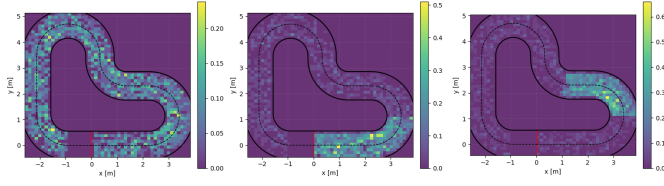


Fig. 5. Three different target-domain off-policy sample distributions used in experiment 6.2. The brighter area stands for states sampled with higher frequency. Left (The whole track is randomly sampled); Middle (target-domain samples biasing round the track's starting point); Right (target-domain samples biasing round the track's mid point)

6.3 Low-Speed-to-High-Speed Transfer

In this section, we demonstrate our method's effectiveness in an exceptionally challenging transfer learning scenario. The expert policy is a high-speed MPCC-CONV racing controller, while the target-domain dataset is collected using a conservative, low-speed PID controller. The discriminative information is chosen to be full states x . The objective is to train a high-speed agent capable of racing effectively in the target domain.

This setting presents two key challenges: (1) a significant distributional discrepancy between $\mathcal{B}_s$ and $\mathcal{B}_t$, arising from the distinct trajectory characteristics of the MPCC-CONV Liniger et al. (2014) and PID controllers; and (2) the heightened sensitivity of high-speed imitation learning to small prediction errors, which demands precise policy alignment. Our results highlight the ability of the proposed framework to preserve high imitation accuracy and stable performance despite substantial distribution shifts.

7. CONCLUSION

This paper studied visual domain transfer for end-to-end imitation learning under a realistic and challenging setting where target-domain data are strictly off-policy, expert-free, and limited. We first provided a theoretical analysis of vision-based imitation learning under domain shift, showing that the target-domain imitation loss can be upper bounded by the sum of the source-domain loss and

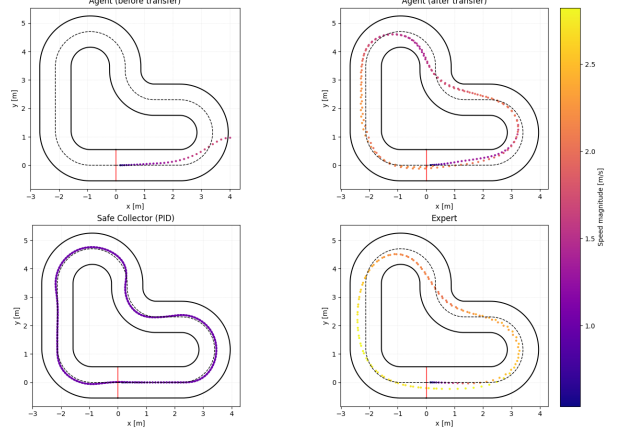


Fig. 6. demonstration of low-speed-to-high-speed transfer.

a state-conditional latent KL divergence between source and target observation models. This result motivates a principled surrogate objective that is both tractable and optimizable from offline data.

Guided by this analysis, we introduced State-Conditional Adversarial Learning (SCAL), an off-policy domain transfer framework that aligns latent representations across domains conditioned on system state. SCAL leverages a discriminator-based estimator of the conditional KL divergence and integrates it with a standard imitation-learning pipeline, enabling expert-free adaptation from a small target-domain buffer paired only with states. Our experiments on visually diverse autonomous driving tasks in the BARC-CARLA environment showed that SCAL achieves robust transfer and strong sample efficiency, often matching or surpassing an idealized DAgger baseline that enjoys significantly stronger target-domain supervision.

Several directions remain open for future work. On the theoretical side, tightening the upper bound and extending the analysis beyond KL-based metrics to other f-divergences or Wasserstein distances may further clarify when SCAL is most effective. On the algorithmic side, improving the stability of adversarial training and exploring alternative density-ratio estimators could enhance robustness in more complex domains. Finally, applying SCAL to real-world systems such as physical autonomous vehicles or aerial robots would provide a definitive test of its practicality under real sensor noise, actuation uncertainty, and safety constraints.

REFERENCES

- Arjovsky, M., Chintala, S., and Bottou, L. (2017). Wasserstein gan. In *Proceedings of the 34th International Conference on Machine Learning*. URL <https://arxiv.org/abs/1701.07875>.
- Arora, S., Zhang, Y., et al. (2018). Games of gan: Game-theoretical models for generative adversarial networks. *arXiv preprint arXiv:1802.05952*.
- Bewley, A., Zemleni, A., Ortenzi, V., and Posner, I. (2018). Learning to drive from simulation without real world labels. *arXiv preprint arXiv:1812.03823*.
- Bojarski, M., Del Testa, D., Dworakowski, D., Firner, B., Flepp, B., Goyal, P., Jackel, L.D., Monfort, M., Muller, U., Zhang, J., et al. (2016). End to end learning for self-driving

- cars. URL <https://arxiv.org/abs/1604.07316>. ArXiv:1604.07316.
- Ganin, Y. and Lempitsky, V. (2016). Domain-adversarial training of neural networks. In *Journal of Machine Learning Research*, volume 17, 1–35.
- Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., and Bengio, Y. (2014). Generative adversarial nets. *Advances in Neural Information Processing Systems*, 27.
- Hansen, N., Jangir, R., Sun, Y., Alenyà, G., Abbeel, P., Efros, A.A., Pinto, L., and Wang, X. (2021). Self-supervised policy adaptation during deployment. In *Proceedings of the 9th International Conference on Learning Representations (ICLR)*. URL <https://openreview.net/forum?id=oV-MjyyGV>.
- Higgins, I., Pal, A., Rusu, A.A., et al. (2017). Darla: Improving zero-shot transfer in reinforcement learning. *arXiv preprint arXiv:1707.08475*.
- Ho, J. and Ermon, S. (2016). Generative adversarial imitation learning. *Advances in neural information processing systems*, 29.
- Kakade, S. and Langford, J. (2002). Approximately optimal approximate reinforcement learning. In *Proceedings of the nineteenth international conference on machine learning*, 267–274.
- Levine, S., Finn, C., Darrell, T., and Abbeel, P. (2016). End-to-end training of deep visuomotor policies. *Journal of Machine Learning Research*, 17(1), 1334–1373.
- Li, B., François-Lavet, V., Doan, T., and Pineau, J. (2021). Domain adversarial reinforcement learning. *arXiv preprint arXiv:2102.07097*. URL <https://arxiv.org/abs/2102.07097>.
- Liniger, A., Domahidi, A., and Morari, M. (2014). Optimization-based autonomous racing of 1:43 scale rc cars. *Optimal Control Applications and Methods*, 36(5), 628–647. doi:10.1002/oca.2123. URL <http://dx.doi.org/10.1002/oca.2123>.
- Long, M., Zhu, H., Wang, J., and Jordan, M.I. (2018). Conditional adversarial domain adaptation. *Advances in Neural Information Processing Systems*, 31.
- Mehta, B., Diaz, M., Golemo, F., Pal, C.J., and Paull, L. (2020). Active domain randomization. In L.P. Kaelbling, D. Kragic, and K. Sugiura (eds.), *Proceedings of the Conference on Robot Learning*, volume 100 of *Proceedings of Machine Learning Research*, 1162–1176. PMLR.
- Mirza, M. and Osindero, S. (2014). Conditional generative adversarial nets. In *arXiv preprint arXiv:1411.1784*.
- Miyato, T. and Koyama, M. (2018). Conditional gans with projection discriminator. *arXiv preprint arXiv:1802.05637*.
- Pfeiffer, C., Wengeler, S., Loquercio, A., and Scaramuzza, D. (2022). Visual attention prediction improves performance of autonomous drone racing agents. *arXiv preprint arXiv:2201.02569*.
- Rahmatizadeh, R., Abolghasemi, P., Bölöni, L., and Levine, S. (2017). Vision-based multi-task manipulation for inexpensive robots using end-to-end learning from demonstration. *arXiv preprint arXiv:1707.02920*.
- Ross, S. and Bagnell, J.A. (2014). Reinforcement and imitation learning via interactive no-regret learning. *CoRR*, abs/1406.5979.
- Ross, S., Gordon, G., and Bagnell, D. (2011). A reduction of imitation learning and structured prediction to no-regret online learning. In G. Gordon, D. Dunson, and M. Dudík (eds.), *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 15 of *Proceedings of Machine Learning Research*, 627–635. PMLR, Fort Lauderdale, FL, USA. URL <https://proceedings.mlr.press/v15/ross11a.html>.
- Stadie, B.C., Abbeel, P., and Sutskever, I. (2017a). Third-person imitation learning. In *International Conference on Learning Representations (ICLR)*. URL <https://arxiv.org/abs/1703.01703>. Preprint.
- Stadie, B.C., Abbeel, P., Sutskever, I., et al. (2017b). A framework for few-shot policy transfer through observation mapping and behavior cloning. *arXiv preprint arXiv:1709.07857*.
- Sun, W., Lim, B., Taylor, M., and Sukthankar, G. (2019). Domain adaptive imitation learning. *arXiv preprint arXiv:1907.03683*.
- Tobin, J., Fong, R., Ray, A., Schneider, J., Zaremba, W., and Abbeel, P. (2017). Domain randomization for transferring deep neural networks from simulation to the real world. In *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 23–30. IEEE.
- Xing, J., Romero, A., Bauersfeld, L., and Scaramuzza, D. (2024). Bootstrapping reinforcement learning with imitation for vision-based agile flight. *arXiv preprint arXiv:2403.12203*.
- Zhang, J. and Cho, K. (2017). Query-efficient imitation learning for end-to-end simulated driving. In *Proceedings of the Thirty-First AAAI Conference on Artificial Intelligence*, 2891–2897. AAAI Press.
- Zhou, X. et al. (2021). Invariance through latent alignment. *arXiv preprint arXiv:2106.10863*.
- Zhu, Y., Joshi, A., Stone, P., and Zhu, Y. (2022). Viola: Imitation learning for vision-based manipulation with object proposal priors. In *Proceedings of Conference on Robot Learning (CoRL)*.

8. APPENDIX

8.1 proof appendix for Lemma 4.1

This section aims to proof the correctness of 4.1

Proof. Note that the expert can be viewed as a history-dependent policy

$$\pi_{\beta} : \bigcup_{k \geq 0} \mathcal{X}^{k+1} \rightarrow \mathcal{U}, \quad u_k^* = \pi_{\beta}(x_{0:k}),$$

For domain $d \in \{s, t\}$, the discounted imitation loss can be framed as

$$\mathcal{J}_d(\theta) = \sum_{k=0}^{\infty} (1 - \gamma) \gamma^k \mathbb{E}_{p^k(y_k, u_k^* | \pi_{\theta})} [\mathcal{L}(\pi_{\theta}(y_k), u_k^*)],$$

where the expert action is $u_k^* = \pi_{\beta}(x_{0:k})$.

We use the following generative model in domain d :

$$\begin{aligned} x_0 &\sim p_0, & y_k &\sim e_d(\cdot | x_k), & l_k &= E(y_k), \\ u_k &= D(l_k), & x_{k+1} &= f(x_k, u_k), & u_k^* &= \pi_{\beta}(x_{0:k}). \end{aligned}$$

At time step k , the imitation loss term can be written as

$$\mathcal{L}(\pi_{\theta}(y_k), u_k^*) = \mathcal{L}(D(l_k), \pi_{\beta}(x_{0:k})).$$

Conditioned on the prefix $x_{0:k}$, the expert action is deterministic, while l_k depends only on x_k . Thus

$$p_d(l_k \mid x_{0:k}, \pi_\theta) = p_d(l_k \mid x_k, \pi_\theta).$$

Define the per-prefix surrogate loss

$$g_d(x_{0:k}) := \mathbb{E}_{l_k \sim p_d(l \mid x_k, \pi_\theta)} [\mathcal{L}(D(l_k), \pi_\beta(x_{0:k}))].$$

Then

$$\mathcal{J}_d(\theta) = \sum_{k=0}^{\infty} (1 - \gamma) \gamma^k \mathbb{E}_{x_{0:k} \sim p_d(x_{0:k} \mid \pi_\theta)} [g_d(x_{0:k})].$$

Using the definition of alignment, the encoder E_ϕ of the agent can induce

$$p_s(l \mid x, \pi_\theta) = p_s(l \mid x, E_\phi) = p_t(l \mid x, E_\phi) = p_t(l \mid x, \pi_\theta),$$

we obtain, for any prefix $x_{0:k}$,

$$\begin{aligned} g_s(x_{0:k}) &= \mathbb{E}_{l_k \sim p_s(l \mid x_k, \pi_\theta)} [\mathcal{L}(D(l_k), \pi_\beta(x_{0:k}))] \\ &= \mathbb{E}_{l_k \sim p_t(l \mid x_k, \pi_\theta)} [\mathcal{L}(D(l_k), \pi_\beta(x_{0:k}))] \\ &=: g(x_{0:k}). \end{aligned}$$

Thus $g_s(x_{0:k}) = g_t(x_{0:k}) = g(x_{0:k})$ for all $x_{0:k}$.

For any x ,

$$\begin{aligned} p_s(u \mid x, \pi_\theta) &= \int \delta(u - D(l)) p_s(l \mid x, \pi_\theta) dl \\ &= \int \delta(u - D(l)) p_t(l \mid x, \pi_\theta) dl \\ &= p_t(u \mid x, \pi_\theta). \end{aligned}$$

Since the two domains share p_0 and the dynamics $x_{k+1} = f(x_k, u_k)$, the Markov chains induced by π_θ are identical. By induction on k , this yields

$$p_s(x_{0:k} \mid \pi_\theta) = p_t(x_{0:k} \mid \pi_\theta), \quad \forall k.$$

Combining $g_s(x_{0:k}) = g_t(x_{0:k}) = g(x_{0:k})$ with $p_s(x_{0:k} \mid \pi_\theta) = p_t(x_{0:k} \mid \pi_\theta)$, we obtain

$$\begin{aligned} \mathcal{J}_s(\theta) &= \sum_{k=0}^{\infty} (1 - \gamma) \gamma^k \mathbb{E}_{x_{0:k} \sim p_s(x_{0:k} \mid \pi_\theta)} [g(x_{0:k})] \\ &= \sum_{k=0}^{\infty} (1 - \gamma) \gamma^k \mathbb{E}_{x_{0:k} \sim p_t(x_{0:k} \mid \pi_\theta)} [g(x_{0:k})] \\ &= \mathcal{J}_t(\theta). \end{aligned}$$

□

8.2 Proof Appendix for Theorem 4.1

For the simplicity of notations, in this proof, we use $p(\cdot)$ as a default shorthand for $p(\cdot \mid \pi_\theta)$ all the visitation distributions. Similarly, we use $p(\cdot \mid x)$ as a short hand for $p(\cdot \mid x, E_\phi)$. That means, by default, we assume every distribution in this proof is conditioned under the agent π_θ . Consider the following derivation

$$\begin{aligned} &|\mathcal{J}_t(\theta) - \mathcal{J}_s(\theta)| \\ &= \left| \mathbb{E}_{p_s(y, u^*)} [\mathcal{L}(\pi_\theta(y), u^*)] - \mathbb{E}_{p_t(y, u^*)} [\mathcal{L}(\pi_\theta(y), u^*)] \right| \\ &= \left| \int \int \mathcal{L}(\pi_\theta(y), u^*) (p_s(y, u^*) - p_t(y, u^*)) dy du^* \right| \\ &\leq \alpha \int \int |p_s(y, u^*) - p_t(y, u^*)| dy du^* \\ &\leq 2\alpha d_{\text{TV}}(p_s(y, u^*), p_t(y, u^*)) \\ &\quad \text{(by the definition of Total Variance)} \\ &\leq 2\alpha \sqrt{\frac{1}{2} d_{\text{KL}}(p_s(y, u^*) \parallel p_t(y, u^*))} \\ &\quad \text{(by Pinsker's Inequality)} \\ &= \alpha \sqrt{2 d_{\text{KL}}(p_s(y, u^*) \parallel p_t(y, u^*))}. \end{aligned}$$

where $\alpha = \sup_{y \in \mathcal{Y}, u^* \in \mathcal{U}} \mathcal{L}(\pi_\theta(y), u^*)$

Then the problem suffices to find an upper bound for $d_{\text{KL}}(p_s(y, u^*) \parallel p_t(y, u^*))$. Note that by the definition γ -discounted distribution and the convexity of d_{KL} . We can obtain the following:

$$\begin{aligned} &d_{\text{KL}}(p_s(y, u^*) \parallel p_t(y, u^*)) \\ &\leq (1 - \gamma) \sum_{k=0}^{\infty} \gamma^k d_{\text{KL}}(p_s^k(y_k, u_k^*) \parallel p_t^k(y_k, u_k^*)) \\ &\leq (1 - \gamma) \sum_{k=0}^{\infty} \gamma^k d_{\text{KL}}(p_s^k(y_k, x_{0:k}) \parallel p_t^k(y_k, x_{0:k})) \end{aligned}$$

To understand why the second inequality holds, recall that the joint distributions $p_d^k(y, u^*) \forall d \in \{s, t\}$ are pushed-forward distributions obtained by applying the function $u^* = \pi_\beta(x_{0:k})$ to the distributions $p_d^k(y, x_{0:k}) \forall d \in \{s, t\}$. Thus, the second inequality holds by the Data Process Theorem.

Since $p_s^k(y_k, x_{0:k}) = e_s(y_k \mid x_k) \cdot p_s^k(x_{0:k})$ and $p_t^k(y_k, x_{0:k}) = e_t(y_k \mid x_k) \cdot p_t^k(x_{0:k})$ by the chain rule of KL divergence, we will have the following:

$$\begin{aligned} &d_{\text{KL}}(p_s^k(y, x_{0:k}) \parallel p_t^k(y, x_{0:k})) \\ &= \mathbb{E}_{p_s^k(x)} [d_{\text{KL}}(e_s(\cdot \mid x) \parallel e_t(\cdot \mid x))] + d_{\text{KL}}(p_s(x_{0:k}) \parallel p_t(x_{0:k})) \end{aligned}$$

Together, we will have the following upper bound for $d_{\text{KL}}(p_s(y, u^*) \parallel p_t(y, u^*))$.

$$\begin{aligned} &d_{\text{KL}}(p_s(y, u^*) \parallel p_t(y, u^*)) \\ &\leq (1 - \gamma) \sum_{k=0}^{\infty} \gamma^k \{ \mathbb{E}_{p_s^k(x)} [d_{\text{KL}}(e_s(\cdot \mid x) \parallel e_t(\cdot \mid x))] \\ &\quad + d_{\text{KL}}(p_s(x_{0:k}) \parallel p_t(x_{0:k})) \} \\ &= (1 - \gamma) \underbrace{\sum_{k=0}^{\infty} \gamma^k \{ \mathbb{E}_{p_s^k(x)} [d_{\text{KL}}(e_s(\cdot \mid x) \parallel e_t(\cdot \mid x))] \}}_{\text{Part A}} \\ &\quad + \underbrace{(1 - \gamma) \sum_{k=0}^{\infty} \gamma^k d_{\text{KL}}(p_s(x_{0:k}) \parallel p_t(x_{0:k}))}_{\text{part B}} \end{aligned}$$

The problem now suffices to find compact bound notations for part A and part B.

8.2.0.1. Part A Based on the definition of γ -discounted distribution, part A can be re-formulated:

$$(1 - \gamma) \sum_{k=0}^{\infty} \gamma^k \{ \mathbb{E}_{p_s^k(x)} [d_{\text{KL}}(e_s(\cdot | x) \| e_t(\cdot | x))] \} \\ = \mathbb{E}_{p_s(x)} [d_{\text{KL}}(e_s(\cdot | x) \| e_t(\cdot | x))]$$

Note that this term depicts the distributional discrepancy determined by the observation model, which is mostly not optimizable. In the following proof, we will refer this term as a constant σ .

8.2.0.2. Part B Consider the following:

$$d_{\text{KL}}(p_s(x_{0:k}) \| p_t(x_{0:k})) \\ = d_{\text{KL}}(p_s^0(x) \| p_t^0(x)) \\ + \sum_{i=0}^{k-1} \mathbb{E}_{p_s^i(x_i)} [d_{\text{KL}}(p_s^i(x_{i+1} | x_i) \| p_t^i(x_{i+1} | x_i))] \\ = \sum_{i=0}^{k-1} \mathbb{E}_{p_s^i(x_i)} [d_{\text{KL}}(p_s^i(x_{i+1} | x_i) \| p_t^i(x_{i+1} | x_i))]$$

The first equality is by the chain rule of KL divergence. The second equality is by the assumption that both domains share the same initial state distribution.

Note that $p_t^i(x_{i+1} | x_i) = p_t^i(x_{i+1} | l) \cdot p_t^i(l | x_i) = \delta(x_{i+1}, D_\theta(l)) p_t^i(l | x_i)$, where δ is Kronecker function. Thus, $p_t^i(x_{i+1} | x_i)$ can be viewed as a distribution obtained by applying channel $\delta(x_{i+1}, D_\theta(l))$ to the distribution $p_t(l | x_i)$. Then, by the Data Process Theorem, we will have $d_{\text{KL}}(p_s^i(x_{i+1} | x_i) \| p_t^i(x_{i+1} | x_i)) \leq d_{\text{KL}}(p_s^i(l | x_i) \| p_t^i(l | x_i)) \forall i, x_i$. With this fact, we can further refine the upper bound for $d_{\text{KL}}(p_s(x_{0:k}) \| p_t(x_{0:k}))$:

$$d_{\text{KL}}(p_s(x_{0:k}) \| p_t(x_{0:k})) \leq \sum_{i=0}^{k-1} \mathbb{E}_{p_s^i(x)} [d_{\text{KL}}(p_s^i(l | x) \| p_t^i(l | x))]$$

Now, plug this back to the expression of Part B, we will get:

$$(1 - \gamma) \sum_{k=0}^{\infty} \gamma^k \{ d_{\text{KL}}(p_s(x_{0:k}) \| p_t(x_{0:k})) \} \\ \leq (1 - \gamma) \sum_{t \geq 0} \gamma^t \sum_{i=0}^{t-1} \mathbb{E}_{p_s^i(x)} [d_{\text{KL}}(p_s^i(l | x) \| p_t^i(l | x))] \\ = (1 - \gamma) \sum_{i=0}^{\infty} \mathbb{E}_{p_s^i(x)} [d_{\text{KL}}(p_s^i(l | x) \| p_t^i(l | x)) \sum_{t \geq i+1} \gamma^t] \\ = (1 - \gamma) \sum_{i=0}^{\infty} \mathbb{E}_{p_s^i(x)} [d_{\text{KL}}(p_s^i(l | x) \| p_t^i(l | x)) \frac{\gamma^{i+1}}{1 - \gamma}] \\ = \sum_{i=0}^{\infty} \gamma^{i+1} \mathbb{E}_{p_s^i(x)} [d_{\text{KL}}(p_s^i(l | x) \| p_t^i(l | x))] \\ = \frac{\gamma}{1 - \gamma} \mathbb{E}_{p_s(x)} [d_{\text{KL}}(p_s(l | x) \| p_t(l | x))]$$

8.2.0.3. Conclusion Putting all the things together, we will get the full upper bound:

$$\mathcal{J}_t(\theta) \leq \mathcal{J}_s(\theta) +$$

$$\alpha \sqrt{\frac{2\gamma}{1 - \gamma}} (\mathbb{E}_{p_s(x|\pi_\theta)} [d_{\text{KL}}(p_s(l | x, \pi_\theta) \| p_t(l | x, \pi_\theta))] + \sigma)$$

where

- $\sigma = \mathbb{E}_{p_s(x)} [d_{\text{KL}}(e_s(\cdot | x) \| e_t(\cdot | x))]$,
- α is the uniform bound over the loss function, with $\alpha = \sup_{y \in \mathcal{Y}, u^* \in \mathcal{U}} \mathcal{L}(\pi_\theta(y), u^*)$.

8.3 Implementation Details for Experiment

Agent Architecture Design The input to the agent consists of an RGB image $y \in \mathbb{R}^{224 \times 224 \times 3}$ and the vehicle's velocity v , forming the observation vector $[y, v]^T$. v is the velocity vector defined as $v = [v_{\text{long}}, v_{\text{tran}}]^T$, where v_{long} is the longitudinal velocity and v_{tran} is the lateral velocity in the vehicle's body frame. The output decisions by both the agent and the expert are $[u_a, u_{\text{steer}}]^T$, corresponding to throttle and steering control. The visual encoder used in our framework is a ResNet-18, which maps the RGB image observation y to a latent vector $l \in \mathbb{R}^{512}$. To balance the dimensionality between the latent vector and the velocity input, the velocity v is first projected into a 16-dimensional space via a linear layer. This transformed velocity vector is then concatenated with the latent vector, resulting in a fused decision vector of dimension 528. This fused vector is passed through a single decision layer to get the output decision.

Discriminator Architecture Design The discriminator is implemented as a two-layer multilayer perceptron (MLP), consisting of a linear layer $\mathbb{R}^{512} \rightarrow \mathbb{R}^{256}$, followed by a ReLU activation layer $\mathbb{R}^{256} \rightarrow \mathbb{R}^{256}$, and a final linear decision layer $\mathbb{R}^{256} \rightarrow \mathbb{R}^1$. The output logits of the discriminator are passed through a sigmoid function.

Gaussian Kernel Estimators We implement $\widehat{p_{\mathcal{B}_t}(x)}$ and $\widehat{p_{\mathcal{B}_s}(x)}$ as two independent Gaussian Kernel Estimators fitted with data from $\mathcal{B}_t$ and $\mathcal{B}_s$ respectively. During adversarial transfer learning, they are fitted only once at the start of the training. Then, they are frozen and treated as two fixed weight functions for KL estimation.

Learning Rates for Transfer Learning One main drawback of adversarial learning families lies in highly sensitive and in-robust learning rates. Finding the right learning rates for the discriminator and the agent usually requires huge efforts of hyper tuning. For different experiments we have done, the learning rate usually vary through a wide range. For the future researchers who want to implement this method, we highly encourage them to carefully tune the agent's and the discriminator's learning rates based on their own problem settings.

State Definition The system state x is defined as: $x = [e_\psi, e_s, K(s_0), K(s_1), K(s_2)]^T$ where e_ψ is the heading error in the Frenet frame, e_s is the deviation from the reference centerline, and $K(s_0), K(s_1), K(s_2)$ denote the curvatures of the reference trajectory at three discretely sampled Frenet coordinates ahead of the vehicle's current position.